



Xilinx's Zynq-7000 applications

ucts. The company claims that its displays are free of lead, mercury and other toxic substances that can harm the environment.

Little chips, big wonders

It is possible for display makers to quickly design and manufacture products with a range of readily-available intellectual property and programmable components.

The field-programmable gate arrays (FPGAs) available from companies like Xilinx and Altera enable faster adoption and migration to the latest technology and standards, interfaces and codecs due to their programmability and flexibility. This allows manufacturers to evolve from 3D TV to Digital Cinema (4K2K) and ultimately to 8K4K Ultra HDTV (such as what was used in theatres in London to broadcast the Olympics footage this year).

"Emerging display technology is an area where FPGAs are always needed, given the newness of the technology. There has been a tremendous amount of innovation and breakthrough in semiconductor and IC technology relative to bringing immersive display technologies such as 3D TV, LED local dimming and 4K2K (Quad HD) to life," says Aaron Behman, senior product marketing manager-Broadcast & Consumer Segment Marketing, Xilinx Inc.

He adds, "The precipitation of mobile devices putting constraints on vid-

eo production and the reach of video networks (to fuel mobile-based insatiable video demands and the like) put the spotlight on FPGAs, which are increasingly the only option to meet the demands of an ever-changing broadcast industry. A growing number of OEMs are adopting these flexible devices because hard-cast silicon solutions like application-specific integrated circuits (ASICs) and application-specific standard products (ASSPs) geared for implementing system integration and video processing,

are too expensive and take too long to develop in an era of rapid standards evolution. The combination of programmable devices, market-specific intellectual property, reference designs and hardware boards fosters broadcast design innovation and algorithm enhancements that deliver real-time capabilities of HD and higher resolutions of Digital Cinema and 4K2K."

The latest 28nm-based FPGA platforms from Xilinx Inc., for example, are aimed specifically at speeding time-to-market, helping both consumer electronics manufacturers and broadcast equipment designers drive up the quality of video, overcome the challenges of processing and moving uncompressed high-definition, 3D and 4K video streams, and lower costs with minimised bill of materials.

As the next innovation wave—4K2K—breaks into the mainstream, delays to market will result in missed revenue opportunities. Xilinx's proven reference designs are geared to shorten time to market for highly differentiated and high-performance visual systems. For instance, Xilinx' Display Targeted Design Platform (TDP) consists of a Kintex-7 FPGA hardware platform with an Acquisition, Contribution, Distribution and Consumption (ACDC) 1.0 baseboard developed by Tokyo Electron Device. The Display TDP enables the stitching together of four 1080i video streams into a seamless 4K2K

image by simultaneously increasing performance while reducing power consumption as compared with previous solutions.

Xilinx' Zynq-7000 family further delivers to customers not just an FPGA or an FPGA with a processor but a truly programmable system-on-chip (SoC) wherein the hardware, software and I/O are each fully programmable. It enables the development of monitors and multiviewers (for example) that require incoming video to be de-interlaced and scaled in real time to support monitor formats up to 4K resolution or in 3D with no loss of video quality. Now multiple windows, 3D graphics for games and immersive ultra-realistic viewing beyond HD, to name a few, are possible—all ultimately changing the way consumers watch and interact with TVs.

Something to look forward to

The display industry is indeed shaping up quite excitingly. Brighter, high-performance, high-definition, low-power, 3D, sun-readable, interactive, immersive—there is just too much happening. And, the trend appears to be on a high!

"There have been a lot of innovations in display over the last several years from electrophoretic displays that are used in eReaders like the Kindle, to new MEMS-based displays like the Qualcomm Mirasol display to OLED displays. All of these technologies will compete against the tried and true LCDs, which will be very hard to displace given the existing investment in manufacturing infrastructure. However, OLED displays are a very competing technology to watch out for, given their power performance and the fact that they can be bent to shape," says Behman. "4K2K displays, or displays that are 4x the resolution of full HD, are also interesting technologies. These displays will offer more detail than full HD. Many models will be hitting shelves this holiday season in the US." ●

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